Exercise 3

Suppose $(X_1, Y_1), \ldots, (X_{n+2}, Y_{n+2}) \in \mathcal{X} \times \mathcal{Y}$ are exchangeable. After observing X_{n+1} and X_{n+2}, select $I \in \{1, 2\}$, where I may depend on both (X_{n+1}, X_{n+2}). Given score $s : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$, define a (full) conformal prediction set $\mathcal{C}(X_{n+I})$ such that

$$P\{Y_{n+I} \in \mathcal{C}(X_{n+I})\} \geqslant 1 - \alpha.$$

Solution The solution is to use a Benferroni correction for the two test points.

Specifically, put $S_i = s(X_i, Y_i)$ and let $S_{(1)} \leqslant \cdots \leqslant S_{(n)}$ be the order statistics for $S_1, \ldots, S_n$. Define

$$k = \left\lceil (n+1)\left(1 - \frac{\alpha}{2}\right)\right\rceil,$$

$$\hat{q} = \begin{cases} S_{(k)}, & k \leqslant n, \\ \infty, & k = n+1, \end{cases}$$

and

$$\mathcal{C}(X_{n+I}) = \{y \in \mathcal{Y} \mid s(X_{n+I}, y) \leqslant \hat{q}\}.$$

Standard results (for example, for full conformal prediction, Theorem 3.2 in the textbook) show that, for $i \in \{1, 2\}$, we have

$$P\{Y_{n+i} \notin \mathcal{C}(X_{n+i})\} \leqslant \frac{\alpha}{2}.$$

Thus:

$$P\{Y_{n+I} \notin \mathcal{C}(X_{n+I})\} = P\left(\bigcup_{i=1}^{2} \{Y_{n+i} \notin \mathcal{C}(X_{n+i})\}\right) \leqslant 2 \times \frac{\alpha}{2} = \alpha.$$

This completes the proof. ////